# Antoine Gauquier

PhD in Computer Science

*Data Management, Machine Learning*

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## Research Interests

My research focuses on **efficient and scalable systems for acquiring and searching large volumes of heterogeneous, real-world data**. My work sits at the intersection of **data management and applied machine learning**, with an emphasis on systems that are **efficient, scalable, and practically deployable**. During my PhD, I have designed and implemented a **focused and RL-based Web crawler** for automated large-scale data acquisition (*EDBT* 2026 ; journal submission under review), a **modular statistical table retrieval system** (STAR) that searches at scale across heterogeneous tabular corpora (*ICDE* 2026 ; extended version in *TLDKS*), and a **structured prediction pipeline for table structure detection and cell role classification** in spreadsheets (*CIKM* 2026). I also built **TabAgree**, a platform for reliable spreadsheet table annotation leveraging inter-annotator agreement (under review). Beyond these main research directions, I have contributed to collaborative work on **heterogeneous graph exploration and efficient entity path finding** (*ADBIS* 2023 ; *ESWC* 2023 ; *Inf. Sys.* 2025), **multimodal information extraction from scientific documents** (*JCDL* 2024 ; *ECIR* 2025), and **uncertainty reasoning and cooperative game theory** (*ECSQARU* 2025).

## Education

2023–2026 **PhD in Computer Science**, *École normale supérieure–PSL*, Paris, France

2022–2023 **M2 in Data Management and Artificial Intelligence**, *Télécom Paris*, Palaiseau, France

2022 **Exchange Semester in Computer Science**, *Université du Québec à Montréal (UQAM)*, Montréal, Canada

2018–2023 **Engineering Diploma (equivalent to M.Sc.) in Computer Science**, *IMT Nord Europe*, Lille, France

## Academic Positions

2023–2026 **PhD Candidate and Teaching Assistant**, *École normale supérieure–PSL*, Paris, France
**Topic :** Efficient and Scalable Methods for the Acquisition, Retrieval, and Understanding of Open Tabular Data.
**Advisors :** Pierre Senellart (Full Professor, *École normale supérieure–PSL*) and Ioana Manolescu (Research Director, *Inria* ; part-time Professor, *École Polytechnique*).
**Committee :** Bernd Amann and Sihem Amer-Yahia (Reviewers) ; Fatiha Saïs (President), Marc Spaniol, and Fabian Suchanek (Examiners).
Defended on September 23, 2026.

Feb 2023–Jul 2023 **Research Intern**, *École normale supérieure–PSL*, Paris, France
**Topic :** Impact of document class on the automatic extraction of mathematical environments from scientific literature.
**Advisor :** Pierre Senellart.

May 2022–Sep 2022 **Research Intern**, *Inria and École Polytechnique*, Palaiseau, France
**Topic :** Path-based ConnectionLens graph exploration.
**Advisor :** Ioana Manolescu.

Sep 2021–Jan 2022 **Part-Time Data Science Assistant**, *Inserm*, Lille, France
Part-time position at the *Translational Research Institute for Diabetes*, carried out alongside academic studies.

May 2021–Sep 2021 **Research Intern in Statistics and Machine Learning**, *CHU de Lille*, Lille, France
**Topic :** Predictive modeling for remote monitoring and long-term outcomes after bariatric surgery.
**Advisors :** Violeta Raverdy (Clinical Researcher, *Translational Research Institute for Diabetes*) and Cristian Preda (Full Professor, *Université de Lille*).

## Publications

### Under Review

Antoine Gauquier, Ioana Manolescu, and Pierre Senellart. TabAgree : An Agreement-Aware Platform for Reliable Spreadsheet Table Annotation. 2026.

Antoine Gauquier, Ioana Manolescu, and Pierre Senellart. Reinforcement Learning–Based Focused Web Crawling for Robust and Scalable Data Acquisition. 2026.

### Peer-Reviewed International Conferences

*Rankings are based on CORE data up to early 2026.*

Antoine Gauquier, Ioana Manolescu, and Pierre Senellart. Structured Prediction for Scalable Spreadsheet Table Understanding: From Cell Types to Table Ranges. *Proceedings of CIKM*, 2026. **[A]**

Antoine Gauquier, Simon Ebel, Helena Galhardas, Théo Galizzi, Ioana Manolescu, Aurélien Peden, and Pierre Senellart. Efficient and Scalable Search for Statistics. *Proceedings of ICDE*, 2026. **[A*]**

Antoine Gauquier, Ioana Manolescu, and Pierre Senellart. Efficient Crawling for Scalable Web Data Acquisition. *Proceedings of EDBT*, 2026. **[A]**

Pratik Karmakar, Antoine Gauquier, and Pierre Senellart. Expected Shapley Value is Shapley Value for Expected Utility Game. *Proceedings of ECSQARU*, 2025. **[C]**

Shrey Mishra, Antoine Gauquier, and Pierre Senellart. Modular Multimodal Machine Learning for Extraction of Theorems and Proofs in Long Scientific Documents. *Proceedings of JCDL*, 2024. **[A*]** *Ranked A* until 2018 (removed—not considered a pure CS conference).*

Antoine Gauquier and Pierre Senellart. Automatically Inferring the Document Class of a Scientific Article. *Proceedings of DocEng*, 2023. **Runner-up for Best Paper Award [B]**

Nelly Barret, Antoine Gauquier, Jia-Jean Law, and Ioana Manolescu. Exploring Heterogeneous Data Graphs Through Their Entity Paths. *Proceedings of ADBIS*, 2023. **[C]**

### Peer-Reviewed International Journals

Antoine Gauquier, Simon Ebel, Helena Galhardas, Théo Galizzi, Ioana Manolescu, Aurélien Peden, and Pierre Senellart. STAR: Efficient, Scalable, and Modular Retrieval of Statistical Tables. *Transactions on Large-Scale Data- and Knowledge-Centered Systems (TLDKS)*, 2026.

Nelly Barret, Antoine Gauquier, Jia-Jean Law, and Ioana Manolescu. Finding meaningful paths in heterogeneous graphs with PathWays. *Information Systems*, 2025. **[A*]** *CORE 2022; Q1 in SJR.*

### Peer-Reviewed International Demonstrations

Shrey Mishra, Neil Sharma, Antoine Gauquier, and Pierre Senellart. TheoremView: A Framework for Extracting Theorem-Like Environments from Raw PDFs. *Proceedings of ECIR*, 2025. **[A]**

Nelly Barret, Antoine Gauquier, Jia-Jean Law, and Ioana Manolescu. PathWays: Entity-Focused Exploration of Heterogeneous Data Graphs. *Proceedings of ESWC*, 2023. **[B]**

### Peer-Reviewed International Workshops

Antoine Gauquier. Towards Efficient Construction of a Traceable, Multimodal, and Heterogeneous Data Warehouse. *Proceedings of VLDB PhD Workshop*, 2024.

Shrey Mishra, Yacine Brihmouche, Théo Delemazure, Antoine Gauquier, and Pierre Senellart. First Steps in Building a Knowledge Base of Mathematical Results. *Proceedings of the 4th Scholarly Document Processing workshop (co-located with ACL)*, 2024.

### Peer-Reviewed National Conferences

Shrey Mishra, Antoine Gauquier, and Pierre Senellart. Apprentissage multimodal modulaire pour l'extraction de théorèmes et de preuves dans des documents scientifiques longs. *Proceedings of EGC*, 2025.

### Thesis, Reports, and Pre-prints

Antoine Gauquier. Efficient and Scalable Methods for the Acquisition, Retrieval, and Understanding of Open Tabular Data. *PhD Thesis, École normale supérieure–PSL*, 2026.

Shrey Mishra, Antoine Gauquier, and Pierre Senellart. Modular Multimodal Machine Learning for Extraction of Theorems and Proofs in Long Scientific Documents (Extended Version). *arXiv abs/2307.09047*, 2024.

Antoine Gauquier. Impact of the document class in the automatic extraction of mathematical environments in the scientific literature. *IMT Nord Europe and École normale supérieure–PSL*, 2023.

Antoine Gauquier. Path-based ConnectionLens graph exploration. *IMT Nord Europe and Inria*, 2022.

## Software, Datasets, and Technology Transfer

*All code and data are released openly, with reproducibility kits reproducing the experiments of the corresponding papers.*

### Technology Transfer

**Radio France**, *StatCheck project*
STAR is integrated into the StatCheck system, used by journalists at Radio France to fact-check claims against INSEE and Eurostat statistics.

### Systems and Software

2026 **STAR–StatCheck**, *GitLab repository*
Open-source Python implementation of the full statistical table discovery pipeline (indexing and natural-language querying), with documentation covering installation, indexing, and querying (*ICDE* 2026 ; *TLDKS* 2026).

2026 **TabAgree**, *GitLab repository*
Multi-annotator platform for spreadsheet table annotation : cell- and table-structure labeling inside a real spreadsheet editor that preserves formatting, and question–table relevance assessment, with inter-annotator agreement reporting. Released with deployment instructions and ready-to-use example data (demonstration paper under review).

2025 **SB-CLASSIFIER Focused Crawler**, *GitHub repository*
Reinforcement-learning focused crawler and all baselines, running on local website replicas (SQLite) or online through Scrapy, with a tool to replicate a live website locally and scripts reproducing all experiments and figures (*EDBT* 2026).

2025 **TRES for Target Retrieval**, *GitHub repository*
Adaptation of the TRES topical focused crawler to dataset acquisition, updated for compatibility with recent deep learning libraries.

### Datasets and Benchmarks

2026 **Statistical Table Discovery Benchmark**, *Zenodo dataset and GitHub reproducibility kit*
7,605 Eurostat tables, two sets of 100 natural-language questions, and nearly 5,000 manually annotated (question, table) pairs : the first such benchmark built on real, varied Open Data. The kit includes the outputs of all compared systems (BM25, STAR, Solo, Pneuma, Birdie) and evaluation scripts.

2026 **StatSheets**, *Zenodo dataset and code*
Benchmark of 737 real-world statistical spreadsheets from 14 public organizations, varied in source, language, topic, size, and format, with 818 tables and more than 7 million cells annotated for cell-type classification and table extraction. Released with the code of all models and, to our knowledge, the only public implementation of SpreadsheetLLM for table extraction, based on Mistral-7B with LoRA (*CIKM* 2026).

## Talks and Presentations

Sep 2026 Efficient and Scalable Methods for the Acquisition, Retrieval, and Understanding of Open Tabular Data, *Ph.D. Thesis Defense*, École normale supérieure–PSL, Paris, France

May 2026 Efficient and Scalable Search for Statistics, *42nd IEEE International Conference on Data Engineering (ICDE)*, Fairmont The Queen Elizabeth, Montreal, Canada

Mar 2026 Efficient Crawling for Scalable Web Data Acquisition, *29th International Conference on Extending Database Technology (EDBT)*, Tampere University, Tampere, Finland

Nov 2025 Efficient Crawling for Scalable Web Data Acquisition, *Valda Team Seminar*, Paris, France

Oct 2025 Efficient and Scalable Search for Statistics, *41st Conference on Databases (BDA) « Gestion de Données – Principes, Technologies et Applications »*, IRIT, Toulouse, France

Sep 2025 Expected Shapley Value is Shapley Value for Expected Utility Game, *18th European Conference on Symbolic and Quantitative Approaches to Reasoning with Uncertainty (ECSQARU)*, Univ. of Hagen, Hagen, Germany

Jan 2025 Modular Multimodal Machine Learning for Extraction of Theorems and Proofs in Long Scientific Documents (Poster), *25th Conference on* Extraction et Gestion des Connaissances *(EGC)*, INSA, Strasbourg, France

Oct 2024 Efficient Crawler for Scalable Web Data Acquisition, *40th Conference on Databases (BDA) « Gestion de Données – Principes, Technologies et Applications »*, CIUR, Orléans, France

Aug 2024 Towards Efficient Construction of a Traceable, Multimodal, and Heterogeneous Data Warehouse, *Prof. Stéphane Bressan's Team Seminar*, NUS School of Computing, Singapore, Singapore

Aug 2024 Towards Efficient Construction of a Traceable, Multimodal, and Heterogeneous Data Warehouse, *50th International Conference on Very Large Databases – PhD Workshop*, Langham Place, Guangzhou, China

Apr 2024 Efficient and Focused Web Crawling for Statistical Data Sources Retrieval, *2nd Workshop « Infox sur Seine »*, Académie du climat, Paris, France

Aug 2023 Automatically Inferring the Document Class of a Scientific Article, *23rd ACM Symposium on Document Engineering (DocEng)*, University of Limerick, Limerick, Ireland

## Research Visits

Jul–Aug 2025 **Visiting PhD Student**, *National University of Singapore and CNRS@CREATE*, Singapore, Singapore

## Teaching Experience

2023–2026 **Teaching Assistant**, *PSL University and Lycée Louis-le-Grand*, Paris, France

- **Introduction to Algorithms** and **Python Programming** (L1) : 106 hours. Laboratory sessions and selected lectures.
- **Differential Calculus** (L2, *PSL University*) : 24 hours in 2024 and 2025. Tutorials.
- **Bibliographic Research** (L2, *PSL University*) : 20 hours in 2023 and 2024. Workshop-style lectures.

## Supervision (Undergraduate Students)

2024 **Léo Boullot**, *PSL University*, Paris, France
Construction of a dataset for the automatic extraction of tabular data from PDF documents.

2024 **Paul Sevestre**, *PSL University*, Paris, France
Automatic semantic interpretation of tabular data.

## Professional Service

### Scientific Community

2025–Present **Program Committee Member**, *EDBT Demonstration Track (2027) ; WWW Research Track (2027) ; VLDB and CIKM Demonstration Tracks (2026) ; WASP (2025 and 2026, co-located with IJCNLP–AACL)*

### Education Community

2024–Present **Selection Committee Member**, *CPES "Data Science, Arts and Cultures"*
Joint program between *PSL* and *Lycée Louis-le-Grand*. Committee responsible for the selection of incoming students.

## Institutional Responsibilities

2025–2026 **PhD Student Representative**, *Board of Directors (Conseil d'Administration), École normale supérieure*

2024–2026 **Representative of Non-Permanent Members**, *Computer Science Laboratory, École normale supérieure*